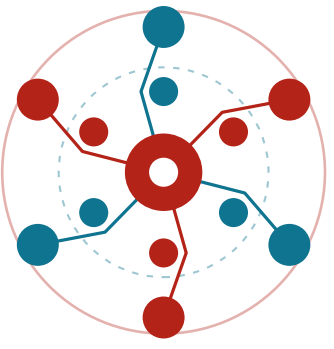
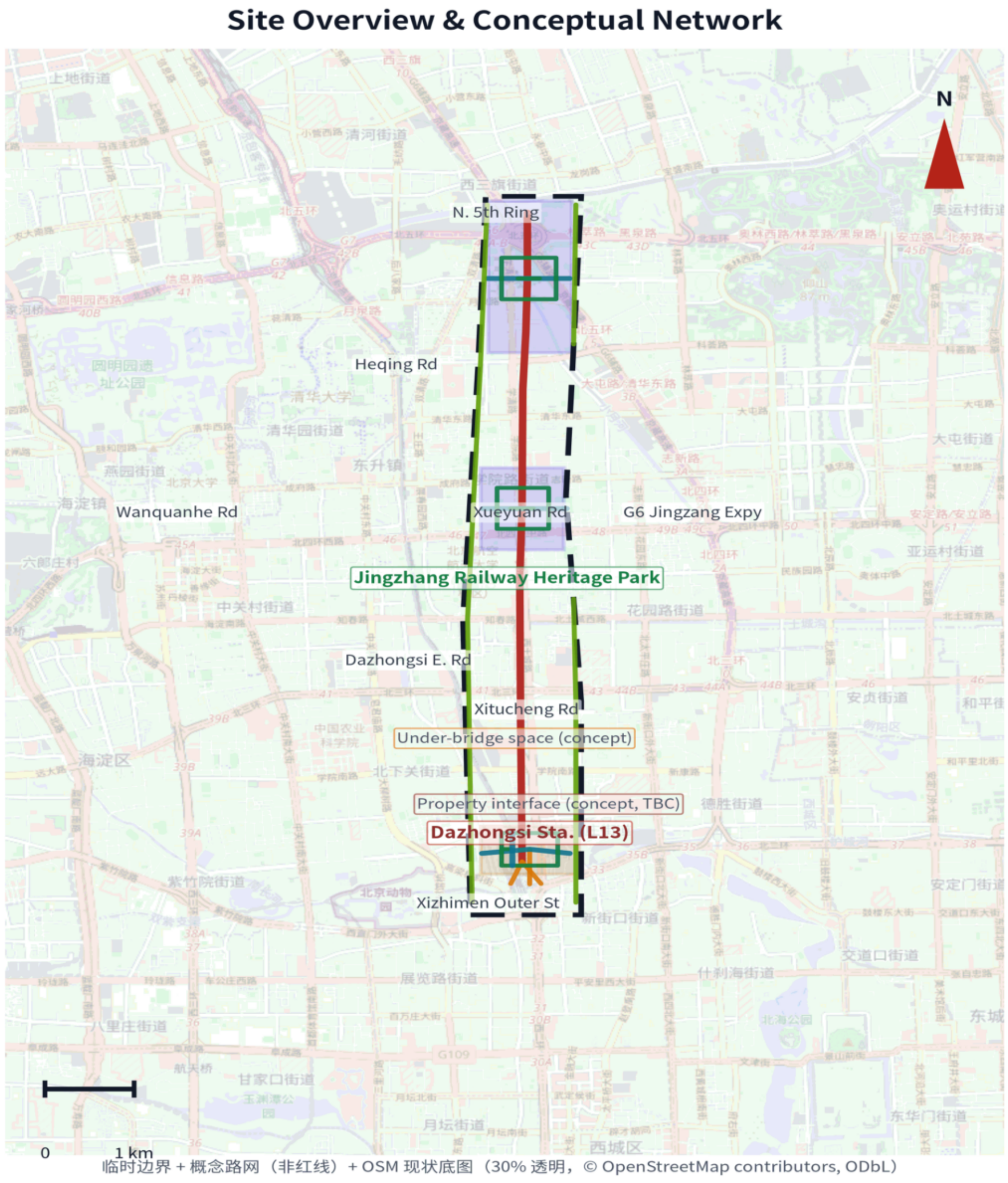




Site overview & conceptual network

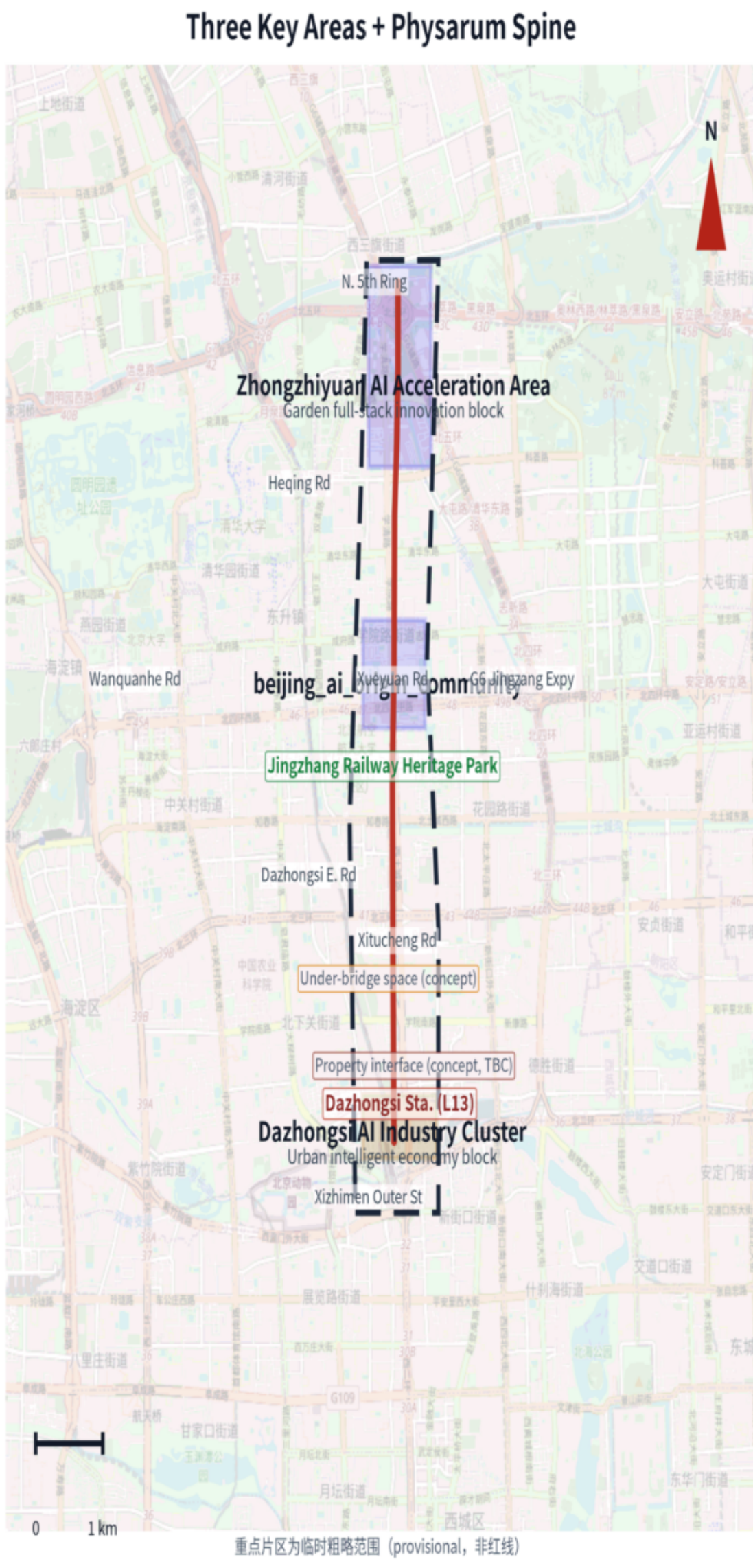


图例 Legend

- 黏菌主静脉 primary vein
- 次级支脉 secondary branch
- 慢行环 local loop
- 京张绿廊 green corridor
- 站点接驳 station feeder
- 众智园片区
- 北京AI原点社区
- 大钟寺AI产业聚集区

边界/路网/片区为算法模拟概念与临时范围（非实测、非审批几何）；道路/站点/公园名称基于 OSM 公开数据与公告文字四至（位置示意）；

Three key areas + Physarum spine

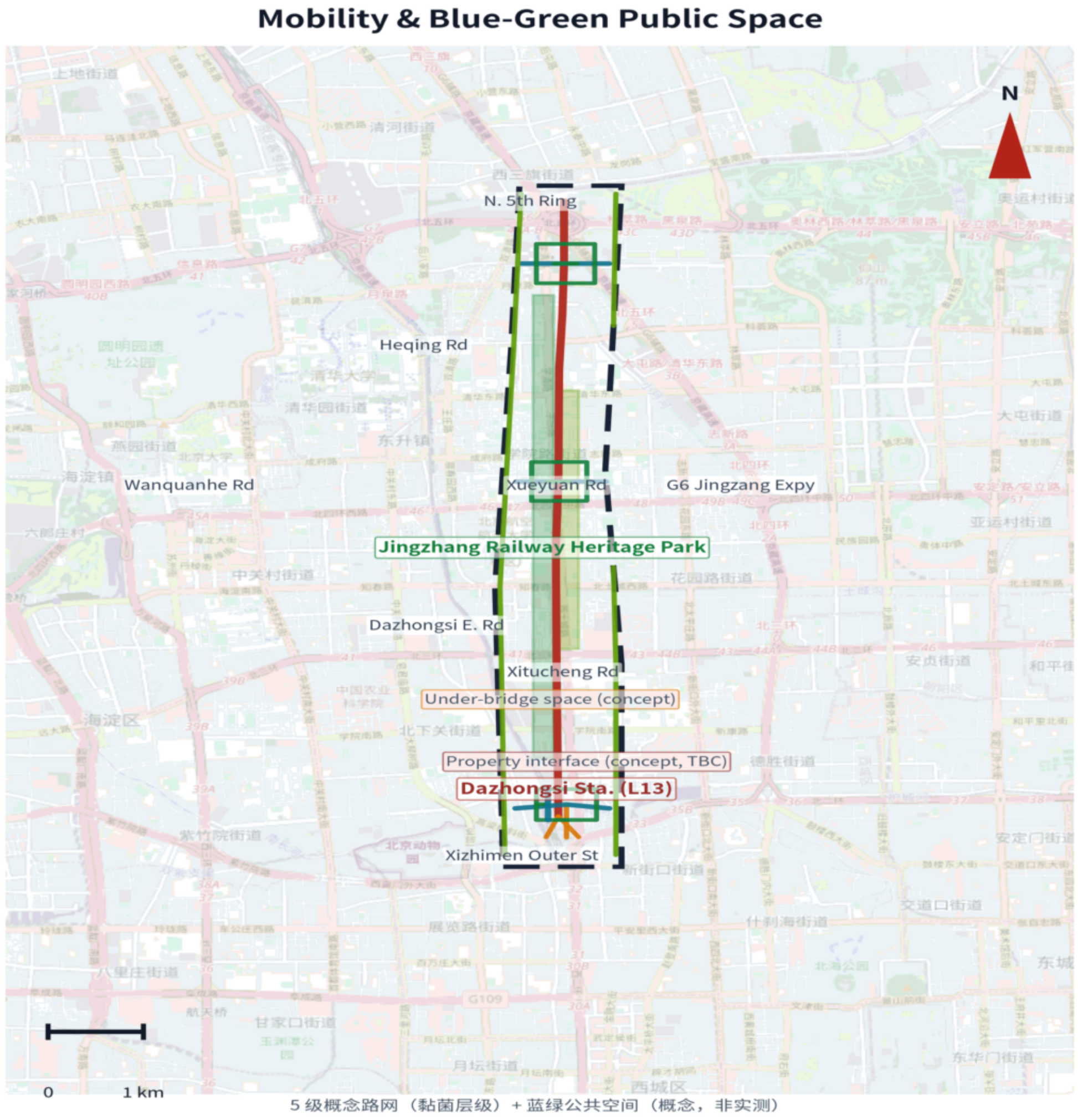


图例 Legend

- 众智园AI自主创新加速区
- 北京AI原点社区
- 大钟寺AI产业聚集区
- 黏菌主静脉 primary vein
- 场地临时边界（虚线）

三处片区边界为公告面积+文字四至推定的临时范围，非官方 polygon，非红线；站点坐标为 OSM 公开数据。

Mobility & blue-green public space

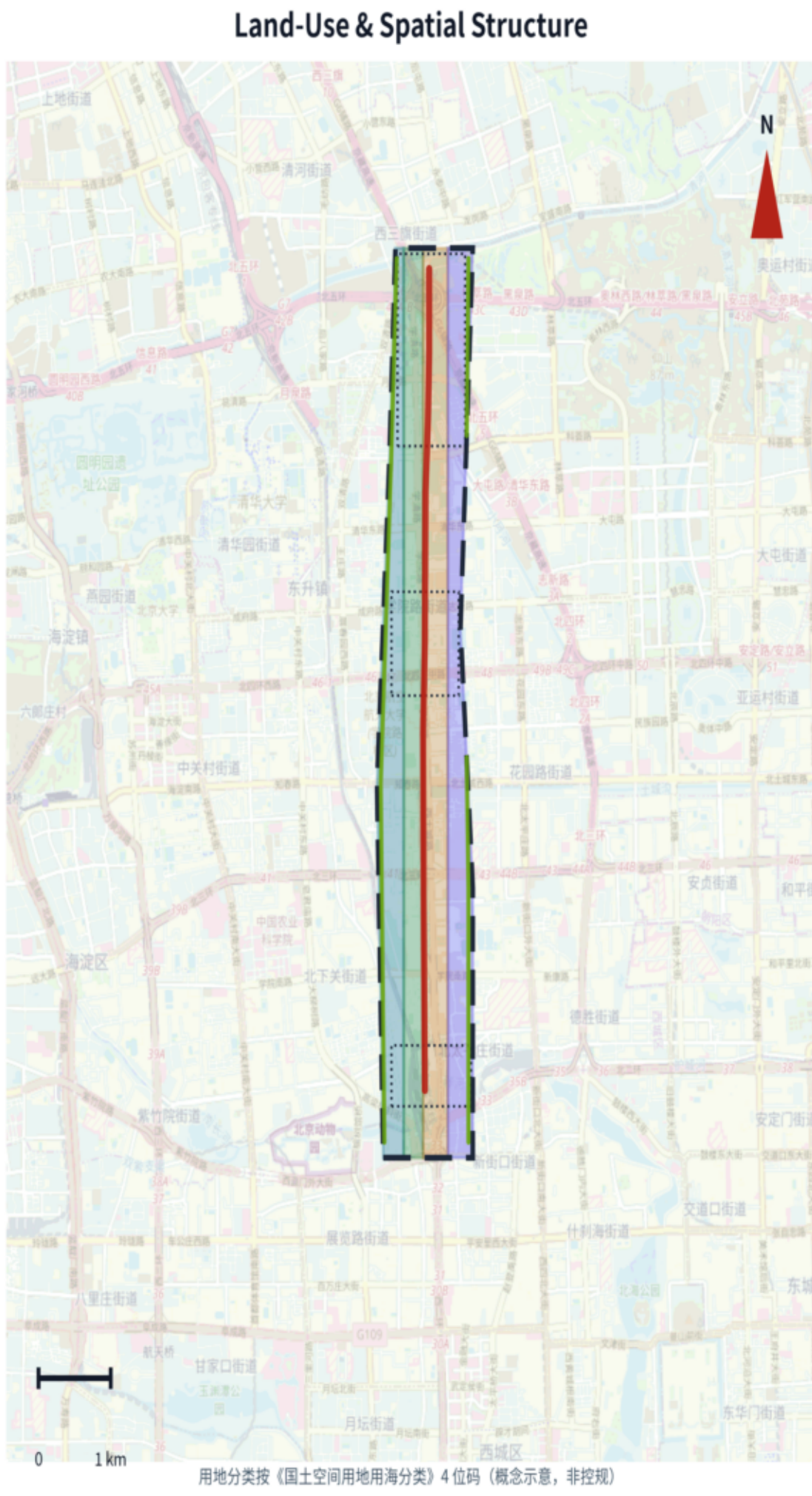


图例 Legend

- 主静脉 primary vein
- 次级支脉 secondary branch
- 慢行环 local loop
- 绿廊 green corridor
- 站点接驳 station feeder
- 公园绿地 green space
- 公共空间 public space

路网为 Bio-Physarum 模拟生成的概念线网（非审批红线、非实测几何）；蓝绿空间为 agent_generated_design 概念示意。

Three-level scope & spatial structure



图例 Legend

- 0802 科研/研发用地 (AI研发试验)
- 1401 公园绿地与开放空间
- 05 商业商务与产业服务
- 0702 社区服务与居住
- 黏菌主静脉
- 京张遗址公园绿廊

用地色块与代码为《国土空间用地海分类》概念示意 (agent_generated_design)，非控规用地，非红线；虚线为三层重点片区临时范围。

Three key areas: Zhongzhiyuan AI innovation acceleration zone (N, ~192.1ha) | Beijing AI Origin community (C, ~104.3ha) | Dazhongsi AI industry cluster (S, ~72.0ha)

Key nodes: Dazhongsi station (Line 13) | Jing-Zhang railway heritage park | under-bridge space (conceptual) | Qinghe innovation interface (conceptual) | university conversion street (conceptual)

Data: network/key areas/boundary are algorithm-generated concepts & provisional ranges (not surveyed, not approval geometry); station coords = OSM public data; heritage protection geometry is unavailable, no redline drawn.

19.20

Optimal eff.

1.143

Baseline

2.802

Run7 frozen

167

Skeleton edges

8813 m

Skeleton length

80.34

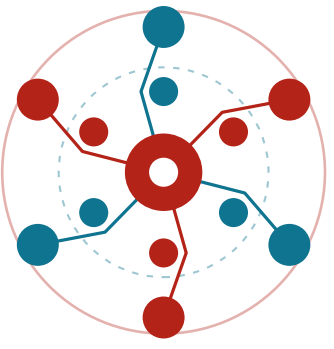
Plan03 UDS

69.1%

Permeable

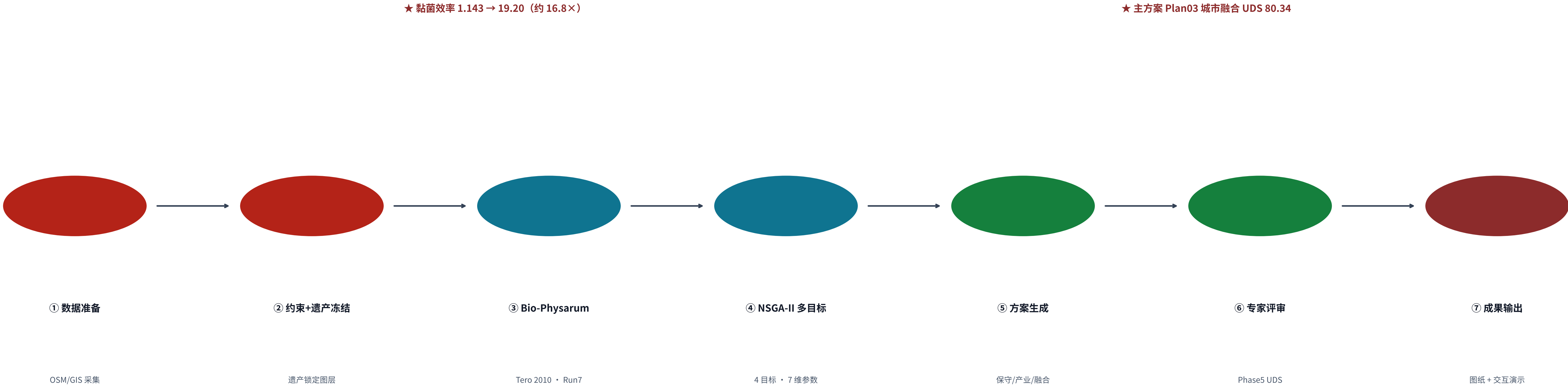
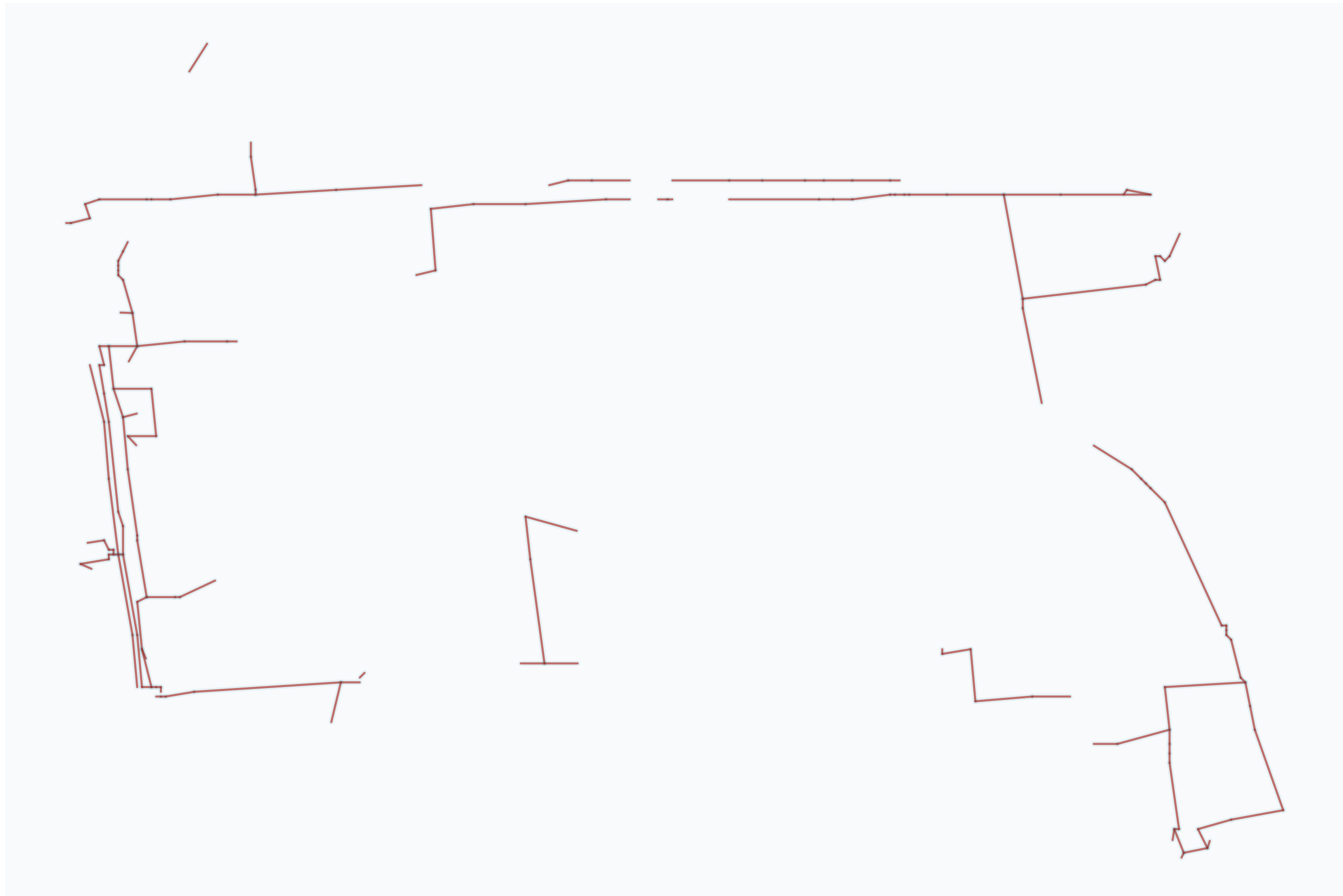
25.2%

Green pen.



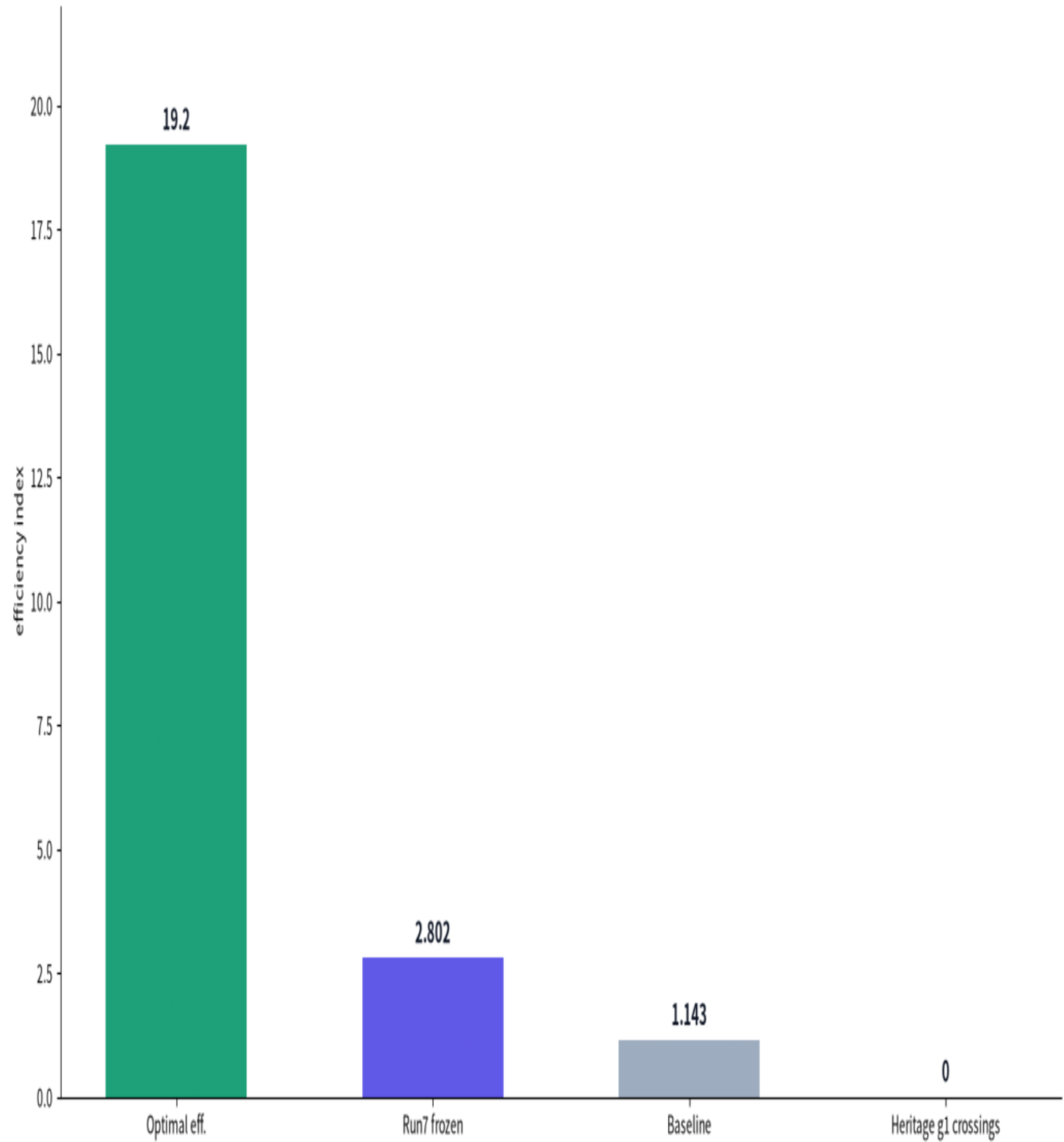
Technical route (Phase 0 → deliverables)

Real Physarum 167-edge skeleton (method validation)



Core metrics & method validation

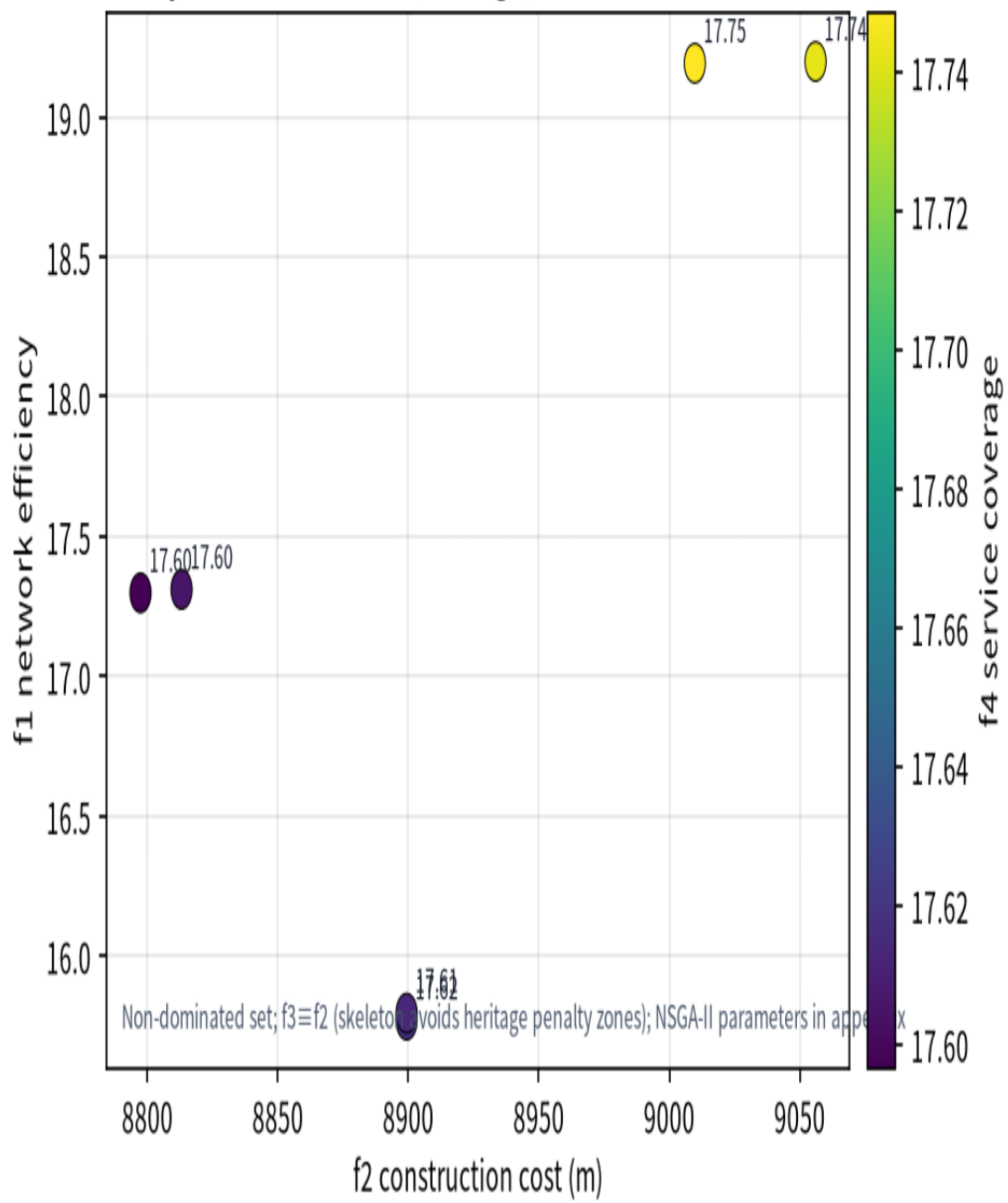
Metrics and method validation



Plan03 urban-integration UDS 80.34 recommended; heritage objective f3≡f2 degenerates to cost — "0 heritage crossings" is g1 hard-constraint behavior only, not an independent site heritage conclusion
Method validation (real Physarum run): 167-edge network / efficiency 19.20 / heritage objective f3≡f2 (degenerates to cost; no independent site heritage conclusion)

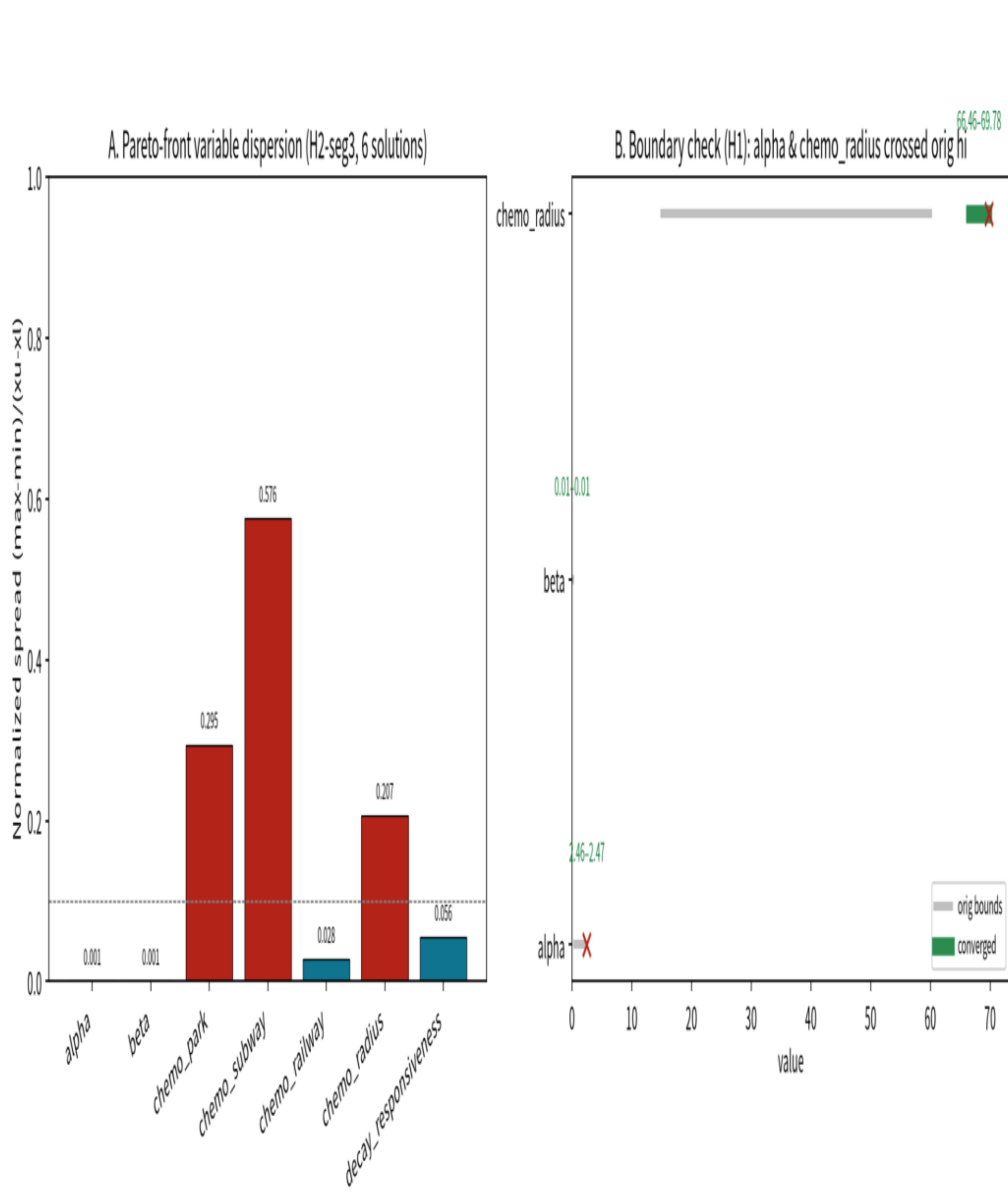
Four-objective Pareto front (6 sol.)

Four-objective Pareto front (H2-seg3, 6 non-dominated solutions)



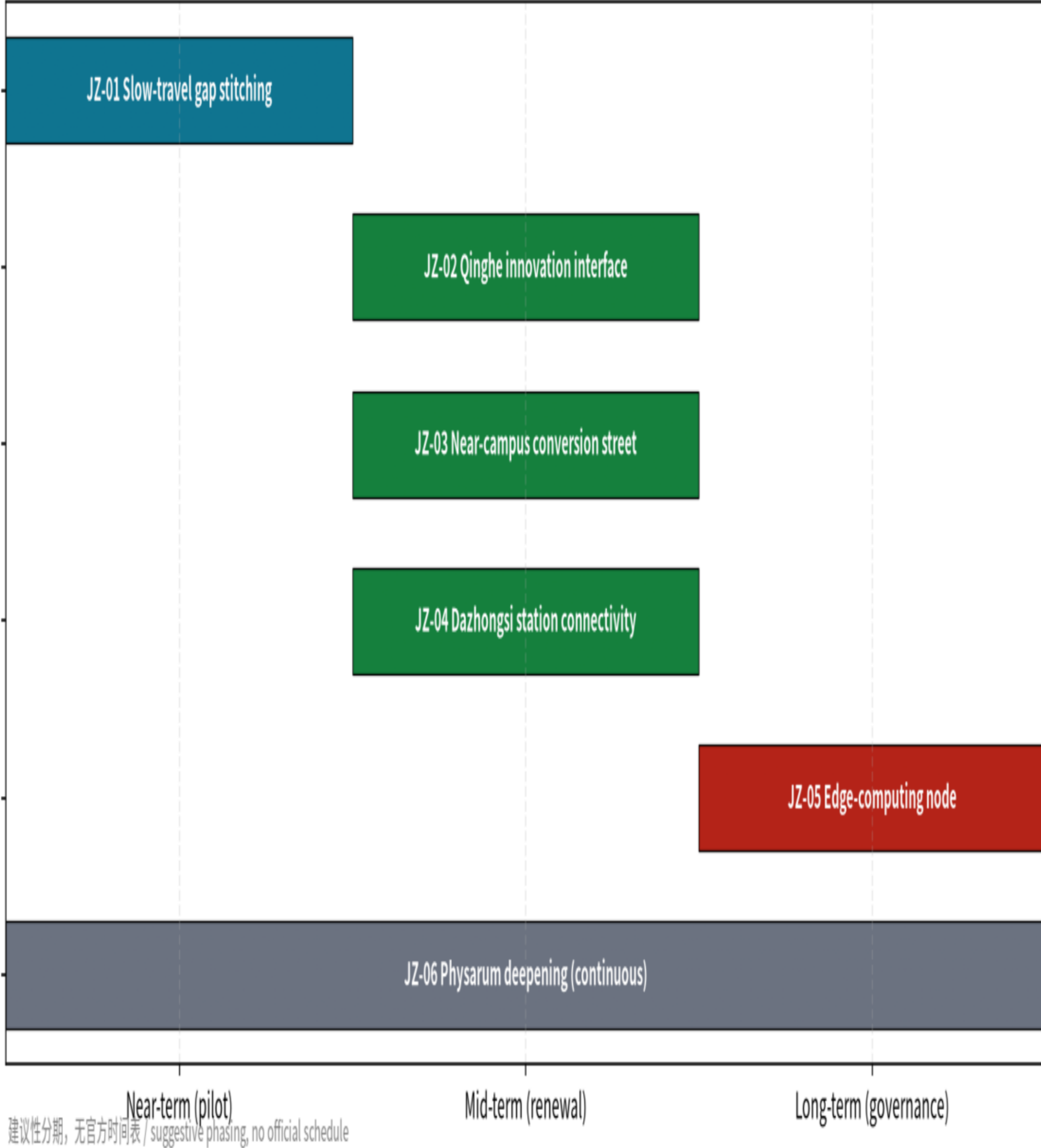
Parameter sensitivity (suggestive)

Parameter sensitivity – Bio-Physarum + NSGA-II (real runs, method validation)



Suggestive phasing gantt (6 projects × 3 phases)

Suggestive phasing gantt (6 projects x 3 phases)



19.20

Optimal eff.

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Run7 frozen

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